

MATTHEW CASALI

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PROFESSIONAL SUMMARY

Innovative and creative acoustic engineer with a strong background in sensing technology and deep learning model development. Experienced in applying advanced signal processing and machine learning classification techniques guided by integration of diverse data sources, high-fidelity acoustic modeling, and measurement for robust model training. Proven ability to deliver groundbreaking products, novel research, and impactful solutions for sponsors.

SKILLS

Software: Python, MATLAB, Git, Linux/Unix, FEA (COMSOL), Apple Core Media
Engineering: Transducer Design, Additive Manufacturing, PCB Assembly, AutoCAD
AI & ML: CNN, LSTM, and Deep Learning architecture, SHAP, Training Parallelization, Weights & Biases
Acoustics: Beamforming, Vibration Analysis, Real-time DSP, Signal Classification, Spectrogram Image Processing

EXPERIENCE

Data Analyst and Acoustic Modeler May 2024 - July 2025
Johns Hopkins University Applied Physics Laboratory

- Built and maintained object-oriented MATLAB toolboxes to streamline acoustic modeling workflows for a user base of 40+ engineers for applications such as beamforming, acoustic energy propagation, and environmental modeling.
- Developed high-precision analytical models in MATLAB to create stable modeling workflows and increase sensor system reliability in extreme environments.
- Constructed complex-geometry FEA models with coupled physics in COMSOL Multiphysics to simulate sensor and beamforming performance, providing stakeholders with essential physically accurate simulated data for decision-making.
- Applied deep-learning neural networks and image-processing to classify massive data-sets to enhance signal processing and noise characterization.
- Enhanced underwater acoustic simulations through the integration of diverse data sources, improving predictive accuracy for real-world applications of sensing and communication technologies.

Research and Development Engineer Sept. 2023 - Jan. 2024
X-Wave Innovations, Inc.

- Authored technical proposals and reports resulting in over \$1,000,000 in SBIR funding and contract extensions from Phase I funding to Phase II.
- Designed and built custom hardware for high-power ultrasonic analysis and force measurement, using AutoCAD, additive manufacturing, and PCB assembly and soldering.
- Built a modeling and testing suite for ultrasonic detection of defects in composite materials for the US Air Force, resulting in a US Patent award.
- Evaluated sensor performance for in-house-fabricated ultrasonic transducers, measuring Transmit-Voltage Response curves using Laser Doppler Vibrometry and oscilloscopes.

PROJECTS

DARPA Metamaterial Transducer — FEA, Beamforming, Technical Writing

- Led the development and prototyping of a novel acoustic metamaterial-based transducer for single-element beamforming.
- Developed a modeling suite to optimize metamaterial structure and simulate experimental characterization.
- Presented “Impedance tube method for characterization of one-dimensional electro-momentum coupled materials” at a meeting of the Acoustical Society of America.

Vaudio iOS App — Swift, Metal, Real-time DSP

- Developed a video-to-audio engine with 10 ms software latency using Metal kernels for feature extraction and DSP.
- Developed Metal GPU-accelerated RGB-to-HSI conversion, intensity gradient calculations, and spectrum synthesis.
- Implemented a 16-frame circular audio buffer with thread-safe locking to ensure dropout-free performance.

AI Card-Player — Pytorch, Weights & Biases, RNN architecture

- Trained a hybrid MLP-RNN agent for the card game Palace using human-in-the-loop and self-play data.
- Achieved a 5% win-rate edge over baseline iterations through iterative self-play reinforcement learning.
- Utilized SHAP analysis to quantify feature importance and interpret AI-discovered strategic play styles.

EDUCATION

The University of Texas at Austin 2023
Master of Science, Mechanical Engineering (Acoustics)

University of California - Santa Cruz 2020
Bachelor of Science, Applied Physics